

**GRAPH THEORY:
HOMEWORK 1
BOOK SECTION 1.1.**

ROB TOMPKINS

1. Consider the following four families of graphs:

$$A = \{\text{paths}\}, \quad B = \{\text{cycles}\}, \quad C = \{\text{complete graphs}\}, \quad D = \{\text{bipartite graphs}\}.$$

For each pair of these families, determine all isomorphism classes of graphs that belong to both families.

Solution. We consider each of the six possible pairs.

(a) $A \cap B$: paths and cycles.

A path has two vertices of degree 1 unless it is a trivial path, while every vertex of a cycle has degree 2. Therefore, no graph is both a path and a cycle. Hence

$$A \cap B = \emptyset.$$

(b) $A \cap C$: paths and complete graphs.

The graphs K_1 and K_2 are paths. If $n \geq 3$, every vertex of K_n has degree $n - 1 \geq 2$, whereas a path has vertices of degree at most 2 and has endpoints of degree 1. Thus

$$A \cap C = \{K_1, K_2\}.$$

(c) $A \cap D$: paths and bipartite graphs.

Every path is bipartite. To see this, color the vertices alternately along the path. Adjacent vertices then receive different colors. Therefore

$$A \cap D = A.$$

(d) $B \cap C$: cycles and complete graphs.

The graph K_3 is a cycle of length 3, namely C_3 . For $n \geq 4$, every vertex of K_n has degree at least 3, while every vertex of a cycle has degree 2. Hence

$$B \cap C = \{K_3\} = \{C_3\}.$$

(e) $B \cap D$: cycles and bipartite graphs.

A cycle is bipartite if and only if it has even length. Therefore

$$B \cap D = \{C_{2k} : k \geq 2\}.$$

(f) $C \cap D$: complete graphs and bipartite graphs.

The graphs K_1 and K_2 are bipartite. If $n \geq 3$, K_n contains a triangle, and therefore is not bipartite. Hence

$$C \cap D = \{K_1, K_2\}.$$

□

2. Prove that removing two opposite corner squares from an 8-by-8 checkerboard leaves a subboard that cannot be partitioned into 1-by-2 and 2-by-1 rectangles. Using the same argument, make a general statement about all bipartite graphs.

Hint: An 8-by-8 checkerboard can be represented by a graph. Each square corresponds to a vertex, and two vertices are adjacent if and only if the corresponding squares share a side.

Proof. Color the checkerboard in the usual way with alternating black and white squares. There are initially 32 black squares and 32 white squares.

Opposite corners of an 8-by-8 checkerboard have the same color. Thus, after removing two opposite corner squares, the remaining board contains either

30 black squares and 32 white squares,

or

32 black squares and 30 white squares.

Every 1-by-2 or 2-by-1 rectangle covers two adjacent squares. Since adjacent squares have opposite colors, every such rectangle covers exactly one black square and one white square.

Therefore, any board that can be partitioned into these rectangles must contain the same number of black and white squares. The board obtained by deleting two opposite corners does not, so such a partition is impossible.

In graph-theoretic language, let G be a bipartite graph with partite sets X and Y . A collection of edges covering every vertex exactly once is a perfect matching. Every edge in a bipartite graph contains one vertex from X and one vertex from Y . Hence, if G has a perfect matching, then necessarily

$$|X| = |Y|.$$

Thus, a bipartite graph whose two partite sets have different cardinalities cannot have a perfect matching. $\square$

3. Let G be a simple bipartite graph with n vertices and m edges. Prove that

$$m \leq \frac{n^2}{4}.$$

Hint: Suppose X and Y are partite sets of G with $|X| = r$ and $|Y| = s$. First show that

$$m \leq rs,$$

and use it to prove

$$m \leq \frac{n^2}{4}.$$

Proof. Let X and Y be the two partite sets of G , with

$$|X| = r \quad \text{and} \quad |Y| = s.$$

Since every edge of a bipartite graph joins a vertex of X to a vertex of Y , and since G is simple, there can be at most one edge for each pair $(x, y) \in X \times Y$. Therefore

$$m \leq rs.$$

Since

$$r + s = n,$$

we have

$$(r - s)^2 \geq 0.$$

Expanding gives

$$r^2 - 2rs + s^2 \geq 0.$$

Hence

$$r^2 + 2rs + s^2 \geq 4rs,$$

so

$$(r + s)^2 \geq 4rs.$$

Since $r + s = n$,

$$n^2 \geq 4rs,$$

and therefore

$$rs \leq \frac{n^2}{4}.$$

Combining the two inequalities gives

$$\boxed{m \leq \frac{n^2}{4}}.$$

Equality occurs when the two partite sets are as equal in size as possible and every possible edge between them is present. $\square$

4. Let G be a graph with girth 4 in which every vertex has degree k . Prove that G has at least $2k$ vertices. Determine all such graphs with exactly $2k$ vertices.

Proof. Since G has girth 4, it contains no triangles.

Choose a vertex $v \in V(G)$. Since every vertex has degree k , v has exactly k neighbors. Let

$$N(v) = \{v_1, v_2, \dots, v_k\}.$$

Because G contains no triangles, no two vertices in $N(v)$ can be adjacent. Otherwise, if v_i and v_j were adjacent, then

$$v, v_i, v_j, v$$

would form a 3-cycle.

Now choose $v_1 \in N(v)$. Since $\deg(v_1) = k$, the vertex v_1 has $k - 1$ neighbors other than v . None of these $k - 1$ vertices can lie in $N(v)$, since that would create a triangle containing v .

Thus the graph contains at least

$$1 + k + (k - 1) = 2k$$

vertices. Hence

$$|V(G)| \geq 2k.$$

Now suppose $|V(G)| = 2k$. Define

$$X = N(v)$$

and

$$Y = V(G) \setminus X.$$

Since $|X| = k$ and $|V(G)| = 2k$, we have $|Y| = k$.

There are no edges between vertices of X , since such an edge would create a triangle with v . Each vertex of X has degree k , and there are exactly k vertices in Y . Therefore every vertex of X must be adjacent to every vertex of Y .

Consequently,

$$G \cong K_{k,k}.$$

Conversely, for $k \geq 2$, the complete bipartite graph $K_{k,k}$ is k -regular, has $2k$ vertices, and has girth 4. Therefore the graphs attaining equality are precisely

$$\boxed{K_{k,k}}.$$

□

5. Let P be the Petersen graph. Use the definition of P given in terms of 2-element subsets of a set of size 5 to prove the following.

- (a) P contains a cycle of length 6.
- (b) P contains a cycle of length 8.
- (c) P does not contain a cycle of length 7.

Hint: For contradiction, assume that C is a cycle in P of length 7. First, explain why every edge of P whose endpoints both lie in C must be an edge of C .

Let u be a vertex of C and let v, w be vertices of C farthest from u . Show that there is some vertex

$$x \notin V(C)$$

such that u and v are both adjacent to x , and u and w are both adjacent to x . Conclude that $\{w, v, x\}$ are the vertices of a 3-cycle in P , which contradicts the fact that P is triangle-free.

Solution. Recall that the vertices of the Petersen graph are the 2-element subsets of

$$\{1, 2, 3, 4, 5\},$$

and two vertices are adjacent precisely when the corresponding 2-element subsets are disjoint.

(a) Consider the sequence

$$12, 34, 15, 23, 45, 13, 12.$$

Consecutive 2-element subsets are disjoint:

$$12 \cap 34 = \emptyset,$$

$$34 \cap 15 = \emptyset,$$

$$15 \cap 23 = \emptyset,$$

$$23 \cap 45 = \emptyset,$$

$$45 \cap 13 = \emptyset,$$

$$13 \cap 12 = \emptyset.$$

Thus these six distinct vertices form the cycle

$$12 - 34 - 15 - 23 - 45 - 13 - 12.$$

Hence P contains a cycle of length 6.

(b) Consider

$$12, 34, 15, 23, 14, 25, 13, 45, 12.$$

Again, every pair of consecutive 2-element subsets is disjoint:

$$12 \cap 34 = \emptyset,$$

$$34 \cap 15 = \emptyset,$$

$$15 \cap 23 = \emptyset,$$

$$23 \cap 14 = \emptyset,$$

$$14 \cap 25 = \emptyset,$$

$$25 \cap 13 = \emptyset,$$

$$13 \cap 45 = \emptyset,$$

$$45 \cap 12 = \emptyset.$$

Therefore

$$12 - 34 - 15 - 23 - 14 - 25 - 13 - 45 - 12$$

is an 8-cycle in P .

(c) Suppose, for contradiction, that P contains a 7-cycle C .

First observe that C cannot have a chord. Indeed, a chord of a 7-cycle divides the cycle into two smaller cycles whose lengths sum to 9. Since the Petersen graph has girth 5, neither of these cycles can have length less than 5. This is impossible because two integers each at least 5 cannot sum to 9. Therefore every edge of P whose endpoints lie in C must already be an edge of C .

Since the Petersen graph is 3-regular, each vertex of C has exactly two neighbors on C and exactly one neighbor outside C .

There are 10 vertices in the Petersen graph, so exactly three vertices lie outside C . Let these vertices be

$$x, y, z.$$

The seven vertices of C each have exactly one edge leading outside C . Thus there are seven edges from C to $\{x, y, z\}$.

Since x, y, z each have degree 3, the sum of their degrees is

$$3 + 3 + 3 = 9.$$

If e denotes the number of edges among x, y, z , then those internal edges contribute $2e$ to this degree sum. Hence

$$7 + 2e = 9,$$

so

$$e = 1.$$

Thus among the three vertices outside C , exactly one pair is adjacent.

Let u be a vertex of C , and let v and w be the two vertices of C at distance 3 from u along the cycle. Since the Petersen graph has diameter 2, both u and v must have a common neighbor, and both u and w must have a common neighbor.

These common neighbors cannot lie on C , since C has no chords. Therefore the common neighbors must lie outside C .

But u has only one neighbor outside C . Call it x . Hence x must be a common neighbor of u and v , and also a common neighbor of u and w . Thus

$$xv, xw \in E(P).$$

Since v and w are adjacent on the 7-cycle C , we also have

$$vw \in E(P).$$

Therefore the three vertices

$$v, w, x$$

form a triangle

$$v - x - w - v.$$

This contradicts the fact that the Petersen graph is triangle-free. Therefore the Petersen graph cannot contain a cycle of length 7. $\square$